

2024 AMC 10A Problem 14 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 14

Problem:

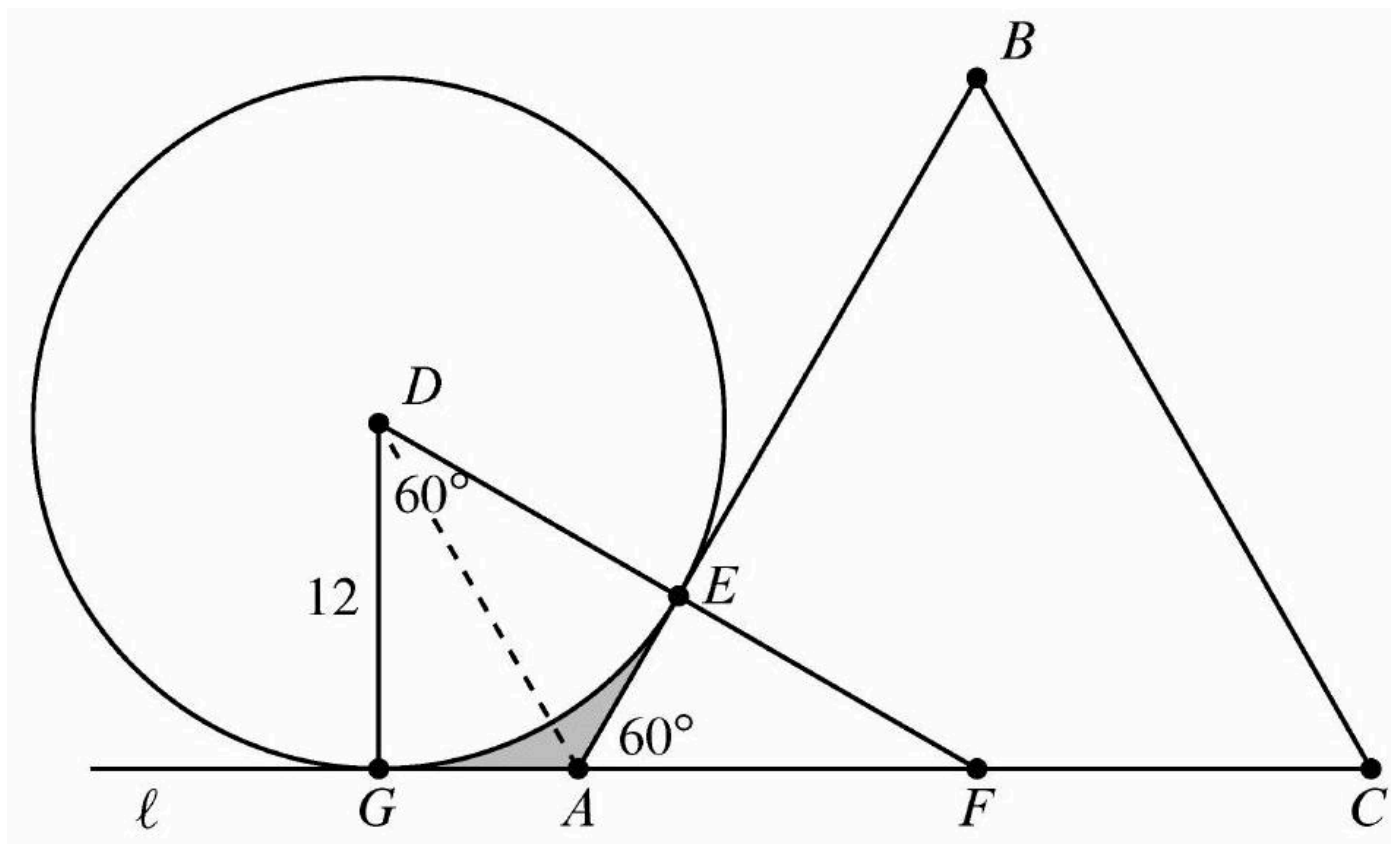
One side of an equilateral triangle of height 24 lies on line ℓ . A circle of radius 12 is tangent to ℓ and is externally tangent to the triangle. The area of the region exterior to the triangle and the circle and bounded by the triangle, the circle, and line ℓ can be written as $a\sqrt{b} - c\pi$, where a , b , and c are positive integers and b is not divisible by the square of any prime. What is $a + b + c$?

Answer Choices:

- A. 72
- B. 73
- C. 74
- D. 75
- E. 76

Solution:

The given situation is shown in the figure below, where D is the center of the circle, E is the point of tangency between the circle and the triangle, F is the intersection of line DE with line ℓ , and G is the projection of D onto ℓ .



Because $\angle BAC = 60^\circ$ and $\angle DEA = \angle DGF = 90^\circ$, both $\triangle AEF$ and $\triangle DGF$ are $30 - 60 - 90^\circ$ right triangles. Therefore $DF = 24$, $EF = 24 - 12 = 12$, $AE = 4\sqrt{3}$, $AF = 8\sqrt{3}$, $FG = 12\sqrt{3}$, and $AG = 12\sqrt{3} - 8\sqrt{3} = 4\sqrt{3}$. The area of kite $GAED$ is twice the area of $\triangle GAD$, so it is $48\sqrt{3}$. The area of the 60° -sector EDG of the circle is $\frac{1}{6} \cdot \pi \cdot 12^2 = 24\pi$. Thus the required area, shaded in the figure, is $48\sqrt{3} - 24\pi$, and the requested sum is $48 + 3 + 24 = \text{(B)} \boxed{75}$.

OR

The required area is $\frac{1}{6}$ of the difference between the area of a circle of radius 12 and a circumscribed regular hexagon. The hexagon is the union of 6 equilateral triangles of side length $s = 12 \cdot \frac{2}{\sqrt{3}}$. The area of the hexagon is $6 \cdot \frac{\sqrt{3}}{4} s^2$, which equals $288\sqrt{3}$. The area of the circle is 144π . Then $\frac{1}{6}$ of the difference is $48\sqrt{3} - 24\pi$, and the requested sum is $48 + 3 + 24 = \text{(B)} \boxed{75}$.

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